



Pranjal
Mechanical Engineering
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
DOB: 04/11/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	
Intermediate	CBSE	SRLVM Sr. Sec. School	2021	96.60%
Matriculation	CBSE	SRLVM Sr. Sec. School	2019	98.20%

Pursuing a Minor Degree in **Artificial Intelligence and Data Science** under the Department of C-Minds, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Achieved AIR **1197** in **IIT JEE Advanced** | Placed in **Top 0.3 %** in JEE Mains | Conferred Scholarship of INR **1L** ['21]
- Attained a **perfect SPI** of **10** during the summer semester, securing **AA grade** across all core courses ['23]
- Aced **NDA** exam conducted by **UPSC**, showcasing exceptional proficiency in **maths** and **analytical** skills ['21]

PROFESSIONAL EXPERIENCE

Mercedes-Benz Research & Development India | *GenAI based Recommendation Engine* [May'24 - Jul'24]

- Designed and executed a proof-of-concept project, crafting a **generative AI** solution for **AIOps** by harnessing **LLMs**
- Integrated **OpenAI GPT-4o** model and **Azure AI Search** to perform **NLP** tasks and search through historical incidents
- Expanded solution's utility to incorporate the analysis of **Datadog metrics**, enabling effective **root cause analysis**
- Incorporated recommendation engine & **anomaly detection** in AIOps saving **28 human hours** of a **14-member** team

Defence Research and Development Organisation | *Quantum Cryptography Research* [May'23 - Jul'23]

Awarded **Letter of Recommendation** for exemplary performance and dedicated work during the Internship

- Utilized **Quantum Cryptography** to ensure a secure communication between two parties that is difficult to decrypt
- Created a **Simulink** model in Matlab for the **simulation** of protocol; Did a *Literature review* of 20+ research papers

TECHNICAL PROJECTS

Classifier in neurodegenerative disorders | *Proteomics lab* | *Prof. Srivastava, IIT Bombay* [Jan'24 - Apr'24]

- Employed diverse **spectroscopy** techniques to extract raw proteomics data from an authentic brain tumor sample
- Conducted **exploratory data analysis** and visualization, utilizing **PCA**, **VIF analysis**, **Correlation heatmaps**
- Utilized **machine learning** algorithms to interpret classifiers, enhancing understanding through **predictive analysis**

Architectural Layout Intelligence | *Computer Vision and ML* | *Prof. Vinay Kulkarni* [Jan'24 - Apr'24]

- Improved **architectural layout design process** through advanced classification using **optimal** past layouts
- Conducted EDA** to identify crucial layout features; **Extracted & engineered features** using **OpenCV** & **scikit-learn**
- Offered best recommendations, optimizing the starting point for new designs, significantly **reducing time** and **effort**

Strategized location of Wireless Network Devices | *Optimization* | *Prof. Avinash Bhardwaj* [Jan'24 - Apr'24]

- Formulated** and translated strategic objectives into **mathematical models** for optimizing the placement of WAPs
- Implemented **Voronoi heuristic approach** with **derivative-free optimization**, and simulated using Python coding
- Evaluated the outcomes of our implemented model against existing solutions and assessed **scalability**

Disney+ Hotstar Clone | *Web Development* [Jun '22 - Jul '22]

- Created a **responsive** and **interactive** user interface (UI) on the front-end that emulates the Disney+ Hotstar website
- Refined** search box, implemented **carousel** with infinite scrolling effect, and added hover effects to the movie cards
- Utilized **CSS** to animate movie cards, **enhancing** their visual appeal, and seamlessly integrated **JavaScript**

Invisibility Cloak | *Computer Vision* | *Electronics and Robotics Club, IIT Bombay* [Jun'22]

- Surveyed the literature on the principles of **Computer Vision** like Image Sampling, Quantisation, Morphology etc.
- Extracted static background frame to create a real-time **mask** that makes objects appear **invisible**, like Harry Potter's

Obstacle Maneuvering Omni Bot | *Electronics and Robotics Club, IIT Bombay* [Aug'22]

- Engineered** a **3 wheeled**, well maneuverable omni bot capable of navigating in **10 directions** within the XY plane
- Created **obstacle avoidance algorithm** tailored for microcontroller & **implemented** it in conjunction with motor driver
- Incorporated a **Wi-Fi module** with the **Blynk IoT** application and configured it through C++ based **Arduino IDE**

ENGINEERING EXPOSURE | RESEARCH & ACADEMIC EXPERIENCE

Computational Quantum Chemistry | Prof. Alok Shukla, Physics Department

[Jan'23 - Feb'23]

- Analysed **Hartree-Fock theory** and **correlation approaches** to examine energy of a **quantum many-body system**
- Used C++ based **Quantum Chemistry package, PSI4** to perform calculations on some molecules/clusters

Computational Fluid Dynamics | Prof. Neeraj Kumbhakarna, Mechanical engineering dept. [Mar'23 - Apr'23]

Simulation and in-lab research of flow around a cylinder using mixing length model for **2-D Navier Stokes** equations

- **Devised framework** for turbulent flow modeling to **reduce computation time** by **30%** compared to existing methods
- Discovered three optimized flow patterns within the circular pipe, to **reduce pressure losses** and thus **save energy**

Space Science and Tech. AwaReness Training | Indian Space Research Organisation (ISRO) [Jul '23]

ISRO has envisaged the **START** programme, as an awareness programme in space science and technology

- Explored domains of **space science research** including Planetary science, Astronomy & Astrophysics, Heliophysics

Cosmology & Dark Matter | Summer of Science '23 | Maths and Physics Club, IIT Bombay [May'23 - Jul'23]

- Explored different domains of cosmology, beginning from the Newtonian view and later **cosmological inflation**
- Researched **vacuum energy**, zero-point **quantum fluctuations**, and candidate particles of dark matter

POSITIONS OF RESPONSIBILITY

Institute Technical Convener | Electronics and Robotics Club, IIT Bombay

[Jun'22 - Apr '23]

Worked in an **8 member team** catering to over **9K+** Electronics & Robotics enthusiasts across the institute

- **Organized** Events, Hackathons, Competitions, Group Discussions, Research projects throughout the year
- **Lectured 200+** enthusiasts and covered basics of OpenCV using PyCharm in Code the pixels, a hands-on **Image processing** session followed by an illustratory project, Invisibility Cloak using Color detection and segmentation
- Indulged **700+ participants** and **100+ mentors** in our flagship, a **Wi-Fi Controlled Bot** making competition, XLR8
- **Presented help sessions** for core electrical, mechanical, **Mechatronics** concepts; **Coordinated** soldering session

Undergraduate teaching assistant | CH 107: Quantum Chemistry

[Dec'22 - Mar'23]

- Part of a **30-member** team of Teaching Assistants assisting the professor in conducting the course smoothly
- Conducted tutorials for **50+** students, clearing their course-related doubts and fostering active **problem-solving**

Winter of Data Science Mentor | Analytics Club, IIT Bombay

[Dec'22 - Jan'23]

- **Led** a team of **4** sophomores and **handled** all the **logistics** in building a "Document Scanner" project
- Provided mentees with best **resources** to enhance their knowledge about Computer Vision using OpenCV, Numpy
- **Solved** their **doubts** regularly through online meets regarding OpenCV and basic **Machine Learning** concepts

JEE Mentor | Physics Wallah

[May'22 - Jul'23]

Physics Wallah is an online ed-tech platform that provides a comprehensive learning experience to **9M+** students

- **Career counseling** and **mentoring** students to help them avoid stressful academic burnouts during JEE times
- **Promoting** betterment of mental health by conducting sessions and enhancing my **communication skills**

TECHNICAL SKILLS

- **Languages & Libraries** : C++, Python, MATLAB, Tensorflow, PyTorch, OpenCV, HTML, JavaScript, $\text{\LaTeX}$
- **Applications** : PyCharm, SolidWorks, Ansys, Arduino, Blender, Eagle, Autodesk, Microsoft Office

RELEVANT COURSES UNDERTAKEN

Mathematics	Calculus I and II, Linear Algebra, Differential Equations, Numerical Analysis
Computers science	Computer Programming, Machine Learning, Speech and NLP, Web Design
Mechanical Engineering	Control Systems, Engineering Mechanics, FEM, Thermodynamics, Structural Materials, Manufacturing Processes, Measurements, CAD, CAM

EXTRA-CURRICULAR ACTIVITIES

Entrepreneurship and Corporate	• Interned at Coding Ninjas in PR and Marketing, and contributed to Project management • Secured funding exceeding 10 lakh rupees for the Tinkerers' Lab, 24/7 lab for engineers • Worked as Campus Ambassador for Unacademy to augment its reach in the campus
YouTube	• Interviewed on Sankalp Bharat's YouTube channel with 700K+ subscribers
Social Work	• Conducted a 3 week bootcamp for under-privileged students; Encouraged STEM education
Event Management	• Led an IT/Software internship prep program for 2,000+ aspiring interns at IIT Bombay • Hosted Career Assistance session in the domain of University Research Internships • Conducted awards ceremony for Institute Technical Projects showcasing 45+ projects